Computational Prototype P0 for Materialization: Calibration, Verdict Maps, and Engineering Readiness

Technology Program Draft

Draft skeleton

Abstract

This manuscript is the canonical computational-paper skeleton for the materialization technology program. It organizes the P0 prototype bundle, the lower theorem-chain calibration, and the verdict map into a clean computational narrative that will later support live code import and Lean-linked experiment definitions.

Contents

1 Scope and Numerical Status **1**

2 Prototype Boundary and Data Contract **1**

3 Calibration Layer **2**

4 Reservoir, Growth, and Observer Layers **2**

5 Primary Sweep Campaigns **2**

6 Verdict Map and Interpretation **2**

7 Bridge to Physical Protocols **2**

8 Reproducibility Contract **2**

9 Limitations and Non-Claims **2**

10 Forward Dependencies **2**

1 Scope and Numerical Status

The computational line currently centers on the P0 prototype. Its role is to operationalize the lowest engineering-ready level of the materialization program and to separate reservoir failure, growth failure, soft completion, glass capture, and hard completion.

2 Prototype Boundary and Data Contract

This chapter will describe the P0 package, the one-run data model, and the meaning of the canonical outputs required for later normalization into a live source tree.

3 Calibration Layer

This chapter will describe the calibration split, the threshold q_*, and the crystalline functional used in the lower theorem chain.

4 Reservoir, Growth, and Observer Layers

This chapter will organize the target compiler, reservoir and seed manager, growth engine, observer diagnostics, functional evaluator, escape estimator, and verdict engine into one computational architecture.

5 Primary Sweep Campaigns

This chapter will present the seed, capacity, compatibility, and metastability sweeps as the first four experimental series of the program.

6 Verdict Map and Interpretation

This chapter will normalize the meaning of `FAIL_RESERVOIR`, `FAIL_GROWTH`, `SOFT_SUCCESS`, `GLASS_CAPTURE`, and `HARD_SUCCESS`.

7 Bridge to Physical Protocols

This chapter will explain how the P0 verdict map feeds post-P0 growth, validation, and metastability protocols for physical-device experimentation.

8 Reproducibility Contract

This chapter will define the canonical import target for the P0 code, the expected tests, and the report artifacts required for a normalized software package.

9 Limitations and Non-Claims

This chapter will make explicit that P0 establishes only the first computationally meaningful lower theorem-chain checkpoint and does not yet generalize to the higher hierarchy levels.

10 Forward Dependencies

This chapter will list the live-source import tasks, the first Lean statement inventory items, and the higher-level calibration tasks that depend on the P0 computational line.